KrishiDisha: Revolutionizing Agriculture with Intelligent Recommendations, Disease Detection, and Yield Prediction

Project Work-II

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

Abhishek Chourasia (EN23CS3T1013)
Goutam Mandloi (EN23CS3T1016)
Shivpratap Singh Panwar (EN23CS3T1017)

Under the Guidance of
Prof. Priyanka Dhasal & Radha Yadav

Department of Computer Science & Engineering
Faculty of Engineering

MEDI-CAPS UNIVERSITY, INDORE- 453331

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INTRODUCTION

Agriculture remains the backbone of global economies, contributing to food security, employment, and economic growth. However, farmers face multiple challenges, including climate variability, declining soil fertility, pest infestations, and inefficient resource allocation. Addressing these issues requires advanced technological solutions that enhance productivity, optimize resource use, and promote sustainability. The integration of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) into agriculture has transformed traditional farming practices into data-driven, precision-based methodologies.

The Fourth Industrial Revolution has driven ML and DL adoption in agriculture, enabling predictive modeling, automated diagnostics, and real-time decision-making. Gradient Boosting Regressor (GBR)-based crop yield prediction models have demonstrated high accuracy in forecasting agricultural output [1], while data preprocessing techniques like StandardScaler and OneHotEncoder have significantly improved model performance in crop yield estimation [2]. Additionally, Random Forest (RF)-based fertilizer recommendation systems have enhanced nutrient application efficiency, reducing environmental waste and improving soil health [3], [16]. Deep learning models, particularly Convolutional Neural Networks (CNNs), have achieved over 95% accuracy in plant disease detection, reinforcing the effectiveness of image-based diagnostics for early disease intervention [4], [18].

Cloud-based decision support systems have further advanced agricultural efficiency by integrating real-time weather data, pest alerts, and crop recommendations, enabling farmers to make timely, data-driven decisions [5]. The adoption of mobile-based platforms has improved farmer engagement, offering instant access to AI-powered insights, resource management strategies, and market trends [6]. Moreover, user-friendly interfaces have been highlighted as critical for widespread adoption, particularly in rural areas with limited technological exposure [7], [20]. Hybrid ML models combining soil and climate data have further personalized crop recommendations, enhancing yield consistency and sustainability [9], [21]. Furthermore, AI and IoT-driven smart farming solutions have optimized agricultural inputs and resource distribution, reducing inefficiencies and boosting productivity [19].

Historical research from 2014 to 2018 laid the foundation for modern AI-driven agricultural advancements. Predictive models combining weather data and soil properties have significantly improved crop yield forecasting [11], while early ML-based image processing for pest detection marked the introduction of deep learning in agricultural pest management [12]. Decision tree-based models for fertilizer optimization have enhanced soil health management and crop nutrition strategies [13]. Additionally, ensemble learning techniques have strengthened crop classification methodologies, contributing to the development of intelligent recommendation systems [14].

While these innovations have significantly advanced precision agriculture, existing AI-driven solutions primarily address individual challenges rather than offering a unified decision-support system. KrishiDisha bridges this gap by integrating multiple AI-driven functionalities into a single, intelligent platform, providing crop recommendations, fertilizer optimization, disease detection, and yield prediction. By employing RF for crop and fertilizer recommendations [3], [9], [21], CNN for disease detection [4], [8], [18], and GBR for yield forecasting [1], KrishiDisha delivers comprehensive, real-time decision-making support to farmers. Additionally, its multilingual accessibility, automated reporting, and cloud-based insights ensure that farmers, regardless of their technological proficiency, can harness AI-driven agricultural solutions for improved productivity and sustainability.

KrishiDisha represents a next-generation AI-powered decision-support system for precision agriculture, seamlessly integrating predictive analytics, real-time diagnostics, and intelligent recommendations. Unlike conventional systems that focus on isolated functionalities, KrishiDisha provides a scalable, inclusive, and holistic agricultural solution, empowering farmers to maximize yield, optimize resources, and drive sustainable farming practices.

LITERATURE REVIEW

The integration of Machine Learning (ML) and Deep Learning (DL) in agriculture has significantly advanced farming practices, addressing challenges such as yield prediction, crop recommendations, fertilizer optimization, and disease detection. This section reviews key research contributions from 2014 to 2025, highlighting developments in these areas.

Machine Learning for Yield Prediction and Crop Optimization: Cloud-based agricultural platforms have demonstrated the effectiveness of integrating multiple functionalities, including yield prediction, disease detection, and fertilizer optimization. A system utilizing Gradient Boosting Regressor (GBR) achieved high accuracy in yield forecasting by analyzing weather patterns, soil conditions, and historical data, showcasing the role of cloud computing in precision farming [1]. Moreover, data preprocessing techniques, such as StandardScaler and OneHotEncoder, significantly improved model performance in yield prediction, emphasizing the importance of efficient data handling in agricultural ML applications [2]. Further advancements in ensemble learning have contributed to improved classification accuracy in crop yield prediction, demonstrating the efficiency of Random Forest (RF) models in agricultural datasets [15].

Fertilizer Optimization and Sustainable Agriculture: ML-based approaches have improved fertilizer management by optimizing nutrient recommendations. An RF-based fertilizer recommendation system effectively optimized fertilizer usage based on soil health and crop requirements, reducing excessive application and minimizing environmental impact [3]. Similarly, ML models have been instrumental in sustainable agricultural practices by enhancing resource utilization and promoting environmentally friendly fertilization methods [16]. Conference studies have also highlighted the benefits of AI-driven fertilizer optimization techniques, proving their effectiveness in increasing productivity while minimizing environmental harm [17].

Disease Detection Using Deep Learning: Deep Learning (DL) techniques, particularly Convolutional Neural Networks (CNNs), have been employed for plant disease detection with over 95% accuracy. These image-based models enhance early disease intervention, reducing crop losses by providing reliable, automated diagnostics [4]. The success of CNNs in disease detection underscores the potential of DL in improving plant health monitoring and precision agriculture. Book references also highlight the importance of integrating image processing and deep learning techniques to enhance precision in disease detection [18].

Decision Support Systems for Smart Farming: The adoption of cloud-based platforms has facilitated real-time decision-making in agriculture. Integrated systems combining weather forecasts, disease alerts, and crop recommendations have enabled farmers to make informed decisions, particularly in areas with limited access to agricultural experts [5]. Mobile-based decision support systems have also gained traction, leveraging smartphone applications to provide real-time insights, including weather conditions, pest alerts, and crop management suggestions, thus improving farmer engagement and resource efficiency [6]. Recent studies emphasize the importance of integrating AI and Internet of Things (IoT) solutions to enhance decision support systems in precision agriculture [19].

User-Centric Agricultural Platforms: For agricultural technologies to achieve widespread adoption, they must be user-friendly and accessible. Studies emphasize the importance of intuitive interfaces in ensuring that farmers, particularly those in rural areas with varying levels of digital literacy, can effectively utilize these tools [7]. Mobile-integrated CNN models for disease classification further demonstrate how AI-powered diagnostics can be made accessible to non-expert users [8]. The integration of web-based agricultural solutions has also contributed significantly to farmer education and technology adoption, as demonstrated by online research contributions [20].

Crop Recommendation Systems and Environmental Factors: Crop recommendation systems integrating environmental factors such as soil quality and climate conditions have enhanced agricultural decision-making. By employing RF-based models, these systems provide tailored recommendations, reducing the risk of crop failure and ensuring higher yield consistency [9]. Additionally, ML-driven fertilizer recommendation models have contributed to sustainable farming by optimizing chemical usage while

improving crop productivity [10]. Hybrid models combining ML with expert-driven agricultural knowledge further enhance crop recommendation accuracy, as explored in various conference proceedings [21].

Foundational Research in Agricultural: AI Earlier studies laid the groundwork for contemporary AI-driven agricultural advancements. The integration of weather data, soil properties, and historical trends significantly improved crop yield prediction, demonstrating the feasibility of multi-source predictive modeling [11]. Image processing-based pest detection methods evolved with the introduction of CNNs, enhancing precision and scalability in pest management [12]. Decision tree-based fertilizer optimization models remain fundamental in precision agriculture [13], while ensemble learning techniques for crop classification have improved model accuracy and reliability in agricultural applications [14]. Studies from various books have contributed to the development of theoretical foundations for AI-driven agricultural innovations.[22].

Bridging Gaps in Agricultural AI: The Role of KrishiDisha: Existing AI-driven agricultural solutions primarily address individual challenges, such as yield prediction, crop recommendation, fertilizer optimization, or disease detection [9], [10]. However, the lack of an integrated decision-support system limits their overall impact on precision agriculture.

To overcome these limitations, KrishiDisha incorporates multiple AI-driven functionalities within a single intelligent platform, leveraging Random Forest (RF) for crop and fertilizer recommendations [3], [9], Convolutional Neural Networks (CNNs) for disease detection [4], [8], and Gradient Boosting Regressor (GBR) for yield forecasting [1]. This integration ensures data-driven, real-time decision-making for farmers. Furthermore, KrishiDisha enhances accessibility and usability through: multilingual support [7], automated reporting, a mobile-friendly interface [6], [20], and cloud-based real-time insights [5]. By providing a comprehensive, scalable, and adaptive solution, KrishiDisha bridges the gap between traditional farming practices and AI-driven precision agriculture, enabling farmers to make informed decisions for improved productivity and sustainability.

PROBLEM DEFINITION

Agriculture faces several challenges that significantly impact both productivity and sustainability:

- **Lack of personalized recommendations:** Farmers often depend on generalized advice for selecting crops and fertilizers, which does not account for farm-specific factors such as soil type, climate, and nutrient levels, leading to inefficiencies and suboptimal yields.
- **Delayed disease identification:** Traditional disease detection methods rely on manual inspection and expertise, resulting in slow identification and late interventions. This delay allows diseases to spread and cause considerable crop losses before corrective measures are taken.
- **Difficulty inaccurate yield prediction:** Farmers lack effective tools to predict crop yields with precision, making it difficult to allocate resources efficiently and plan for future crop cycles and market strategies.
- **Limited access to resources in rural areas:** Farmers in remote locations often struggle with limited access to modern agricultural technology, expert advice, and timely insights, hindering their ability to make data-driven decisions for improved crop production.

KrishiDisha addresses these issues by offering an intelligent platform that integrates AI and ML techniques. The platform uses advanced models like Gradient Boosting Regressor for accurate crop yield prediction, providing farmers with timely insights to optimize resource allocation. It also offers personalized crop and fertilizer recommendations, coupled with real-time disease detection through CNNs for image analysis, enabling early disease intervention. Designed for accessibility, KrishiDisha ensures that farmers, particularly in rural areas, have easy access to the resources and insights necessary to improve agricultural productivity, sustainability, and profitability.

OBJECTIVES

- **Efficient Crop and Fertilizer Recommendations:** Build a system that leverages Random Forest to provide accurate crop and fertilizer recommendations tailored to specific farm conditions, such as soil composition, temperature, humidity, and rainfall.
- **AI-powered Disease Detection:** Implement Convolutional Neural Networks (CNNs) to enable real-time detection of crop diseases through image analysis, ensuring timely intervention to minimize crop losses.
- **Precise Crop Yield Prediction:** Use Gradient Boosting Regressor (GBR) for accurate yield predictions based on environmental factors, crop type, and historical data, assisting farmers in resource planning and market strategies.
- **User-Friendly and Multilingual Interface:** Develop an intuitive platform with multilingual support in Hindi, Tamil, Telugu, Punjabi, Kannada, and English, ensuring ease of use for farmers with diverse linguistic backgrounds and technological proficiencies.
- **Resource Optimization and Sustainability:** Promote sustainable farming practices by optimizing the use of fertilizers, reducing environmental impact, and improving resource allocation to maximize agricultural productivity.
- **Integration of Features:** Create a holistic solution combining crop recommendations, fertilizer suggestions, disease detection, and yield prediction within a single platform to address multiple farming challenges.
- **Accessibility for Rural Farmers:** Ensure the platform is accessible on mobile devices, providing rural farmers with real-time insights and tools for decision-making, regardless of their location.
- **Scalability and Adaptability:** Design the system to handle the evolving needs of agriculture, allowing for integration of new crops, diseases, and datasets as technology and farming practices advance.
- **Empowering Farmers Through Technology:** Leverage AI and machine learning to bridge the gap between traditional farming and modern precision agriculture, empowering farmers with actionable insights to improve productivity and profitability.

METHODOLOGY

The KrishiDisha platform was designed and developed following a structured methodology to provide an intelligent and efficient agricultural support system. The methodology focuses on providing a seamless user experience for both farmers and admins while leveraging machine learning models for crop and fertilizer recommendations, crop yield prediction, and crop disease detection. Below are the detailed steps involved in the system's development and operation.

- **Frontend and Backend Development:** The frontend is built using HTML, CSS, and JavaScript for a user-friendly interface, while the backend is powered by Python (Flask) and MySQL for data management and secure processing. The integration ensures seamless interaction between users and the system's features.
- **User Registration and Authentication:** Farmers register by providing their details, which are stored securely in MySQL with hashed passwords. Admin verifies their details via the Admin Dashboard and once verified, farmers gain access to the platform.
- **Farmer Portal and Key Features:** Upon login, farmers access the Farmer Portal, offering functionalities like Crop Recommendations (using Random Forest), Fertilizer Optimization, Crop Disease Detection (via CNN), and Yield Prediction (using Gradient Boosting Regressor). The portal also supports Multilingual Support and Report Generation for a personalized and accessible experience.
- **Machine Learning Model Development:** KrishiDisha employs a Random Forest model for crop and fertilizer recommendations, using feature scaling for better accuracy. A CNN model detects crop diseases based on image analysis, and a Gradient Boosting Regressor predicts crop yields, achieving a high R^2 score of 0.9793, ensuring reliable performance.
- **Additional Features:** The platform provides Multilingual Support and generates downloadable reports summarizing the farmer's inputs and recommendations, enhancing usability and accessibility.
- **Model Deployment and Data Storage:** The trained machine learning models are saved as pipelines for real-time predictions. Data is securely stored in the database for future analysis and report generation.

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